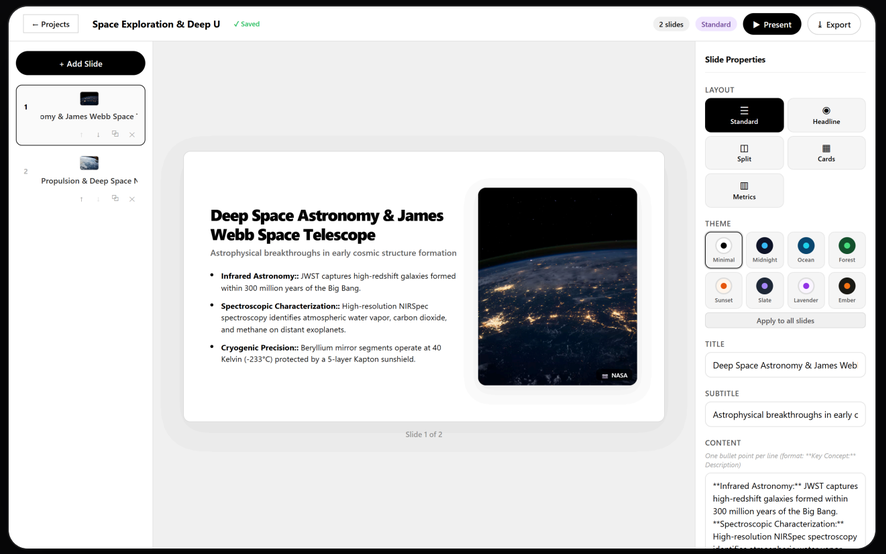
PROGRESS REPORT II — MILESTONE PHASE 2 TO 4

PPT-GEN
Cloud & AI Full-Stack

Full-stack MERN presentation platform powered by Google Gemini 3 Flash, concurrent Unsplash HD imagery, JWT authentication, Phase 4 Present Mode, and automated cloud deployments.



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Under the Guidance of: **Dr. Meenakshi** | Department of Electronics, J.C. Bose UST, YMCA

Live Web: **https://ppt-gen-two.vercel.app** | API: **https://ppt-gen-backend-doal.onrender.com**

Executive Summary & System Evolution

REPORT II

Phase 1 Baseline (Prototype)

Client-Only Architecture

Initial milestone focused strictly on client-side React 19 canvas rendering and in-memory presentation schemas with local state persistence.

Generic Placeholder Images

Early visual previews relied on static CSS watermarks or arbitrary mock placeholders without external content correlation.

Manual Slide Authoring

Required users to write content bullet-by-bullet or use mock templates without live generative artificial intelligence capabilities.

No User Persistence or Cloud State

Decks were isolated to single browser instances with zero authentication or multi-device cloud synchronization.

Phase 2–4 Transformation (Production Full-Stack)

Production Express & MongoDB Atlas Backend

Full RESTful micro-service with BCrypt password hashing, stateless JSON Web Tokens (JWT), and MongoDB Atlas cloud storage.

Google Gemini 3 Flash Prompt Engineering

Sub-2 second structured JSON inference generating 15-25 word factual bullets (****Key Concept:****) with zero marketing fluff.

Concurrent Unsplash HD Visual Pipeline

Automated contextual photo matching with parallel fetching, photographer attribution badges, and 42% split layout image cards.

Phase 4 Present Mode & 5 Streamlined Exports

Fullscreen presentation engine with keyboard controls, auto-hiding HUD, and strictly focused 5 production export formats.

Full-Stack Production Technology Stack

ARCHITECTURE

Tier	Technology & Version	Core Responsibility	Deployment
Frontend SPA	React 19 + Vite 8.2 + React Router v7	Modern concurrent UI, aspect-ratio locked 16:9 canvas, and modular view orchestration.	VERCEL EDGE
Backend API	Node.js v26 + Express 4.19 + CORS	Stateless REST API, JWT auth gatekeeper, Unsplash proxy, and Gemini prompt orchestrator.	RENDER CLOUD
Cloud Database	MongoDB Atlas (Mongoose ODM)	Distributed document store preserving User credentials, deck hierarchies, and slide entities.	ATLAS CLUSTER
Generative AI	Google Gemini 3 Flash Preview	Structured JSON schema synthesis generating high-density technical and domain slide summaries.	GOOGLE AI API
Visual Media CDN	Unsplash REST API (Official)	Concurrent keyword-driven photography retrieval with author attribution and orientation filtering.	UNSPLASH CDN
Export Engines	jsPDF + html2canvas + docx + pptxgenjs	100% client-side binary compilation for 5 production formats: PDF, PPTX, DOCX, PNG, JPEG.	IN-BROWSER
Auth & Security	BCrypt.js + JSONWebToken (JWT)	Salted password hashing (10 rounds) and HMAC-SHA256 bearer token authorization middleware.	ACTIVE

End-to-End System Architecture

DATA FLOW

Vercel React SPA
Landing, Auth, Studio Dashboard, 16:9 Editor, and Present Mode.

→

Render Express API
JWT verification, input validation, and asynchronous microservice coordination.

→

Google Gemini 3
Structured schema synthesis with domain bullets & image query keywords.

→

Unsplash API
Concurrent parallel image retrieval for each generated slide topic.

→

MongoDB Atlas
Atomic persistence of authenticated deck state and slide arrays.

Stateless Microservice Design

Decoupled frontend and backend communicating exclusively over HTTPS with JWT bearer authentication. Frontend assets cache on Vercel's global CDN while the Express backend executes on Render container infrastructure.

Dual Engine Parallel Synthesis

Generative text synthesis and visual asset discovery occur in a synchronized multi-stage pipeline: Gemini produces topic-aligned image search keywords, which trigger concurrent Unsplash queries via Promise.all.

Generative AI & Image Pipeline Workflow PIPELINE

01 INPUT

Prompt & Count

User specifies deck topic and slide count (3 to 15) in the Studio Dashboard AI Modal.

02 LLM

Gemini Synthesis

Gemini 3 Flash returns strictly typed JSON with 15-25 word factual bullets and landscape query tags.

03 PARALLEL

Unsplash Pipeline

Backend executes concurrent Unsplash searches via Promise.all, securing high-res photography and credits.

04 PERSIST

Atlas Database Save

Complete presentation with integrated images and layout metadata commits atomically to MongoDB.

05 RENDER

16:9 Canvas Display

React mounts the presentation immediately into the editor with 42% image cards and present mode triggers.

Prompt Engineering Rule: Gemini is strictly prompted to prohibit marketing buzzwords and produce 15–25 word factual bullets formatted as **Key Concept:** Description with real-world metrics.

Database Models & Authentication Security

DATA SCHEMA

User Model & JWT Authentication

- **name: String (Trimmed, 2–50 characters)**
  Author profile identity displayed on studio dashboard and export metadata.
- **email: String (Unique index, normalized lowercase)**
  Primary authentication credential preventing duplicate registration conflicts.
- **password: String (BCrypt Salted Hash)**
  Pre-save Mongoose hook runs bcrypt.genSalt(10) ensuring raw passwords never touch disk.
- **JWT Session Management**
  Issues HMAC-SHA256 signed bearer tokens with 7-day expiration and automatic 401 redirect protection.

Project & Slide Schema Hierarchy

- **userId: ObjectId (Indexed foreign key)**
  Strict tenant isolation ensuring users only access and modify their own presentation decks.
- **slides: [SlideSubdocument]**
  Sequential collection containing slide id, title, subtitle, layout, and content array.
- **image: { url, alt, photographerName, photographerUrl }**
  Embedded object linking high-res Unsplash photo asset and required API attribution badges.
- **timestamps: { createdAt, updatedAt }**
  Automated change tracking facilitating dashboard recency sorting and audit histories.

Landing Page & Team Attribution

SHOWCASE 1

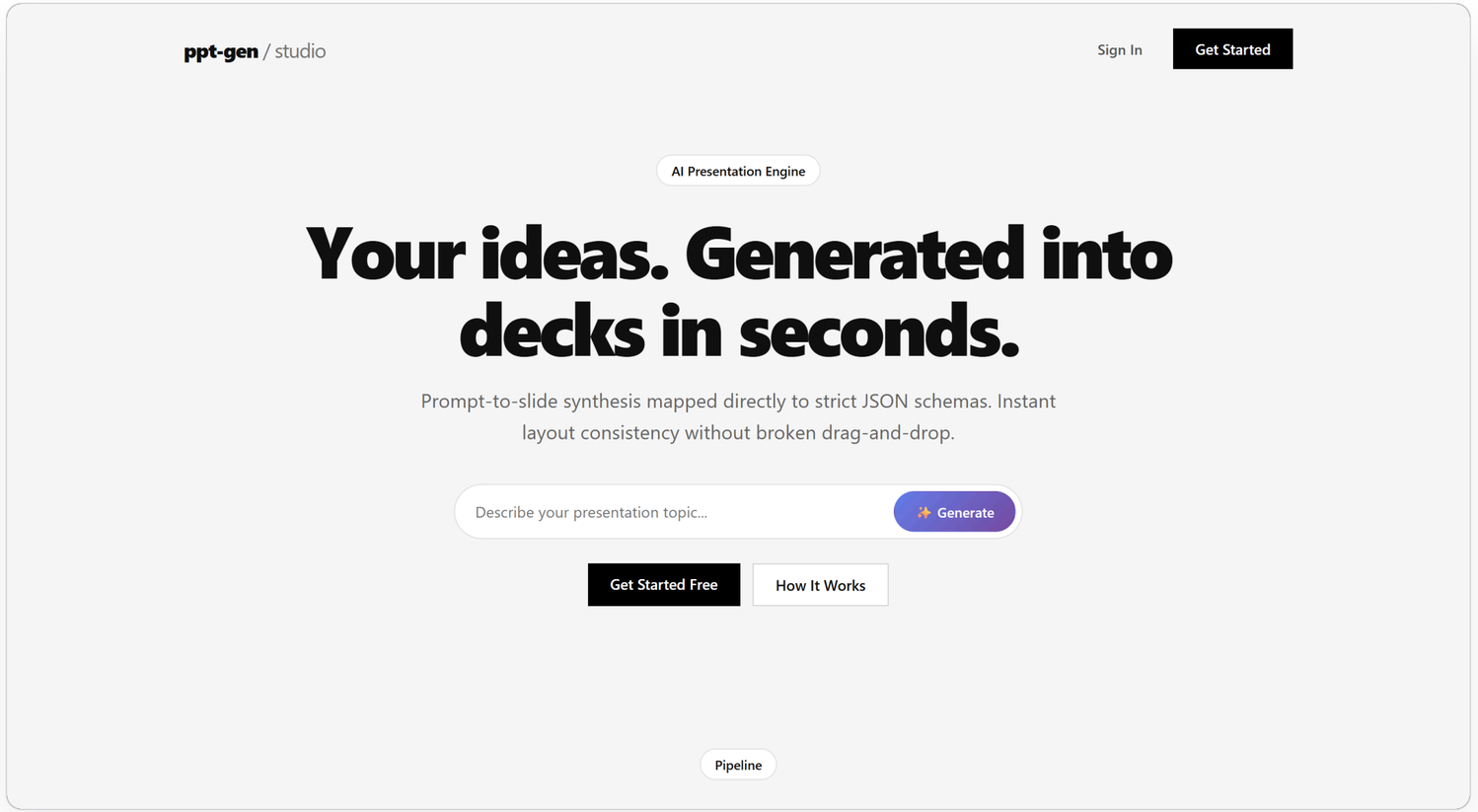


Figure 1: Production landing interface deployed on Vercel CDN

https://ppt-gen-two.vercel.app/

Feature Highlights

- **Hero Value Proposition:** Clean black & white minimalist typography conveying AI-accelerated presentation authoring.
- **Streamlined 5 Exports Pill:** Live badge declaring support for strictly PDF, JPEG, PNG graphic, Word (.docx), and PowerPoint (.pptx).
- **Author Attribution:** Professional footer crediting *Built by Vikrant Tewatia & Sushant*.
- **Direct Contact Hyperlinks:** Integrated mailto: links for vikrtew@gmail.com and sushantbhadana99@gmail.com.
- **Session-Aware Navigation:** Smart navbar toggling between "Get Started" and "Open Studio" based on token state.

JWT User Authentication & Security

SHOWCASE 2

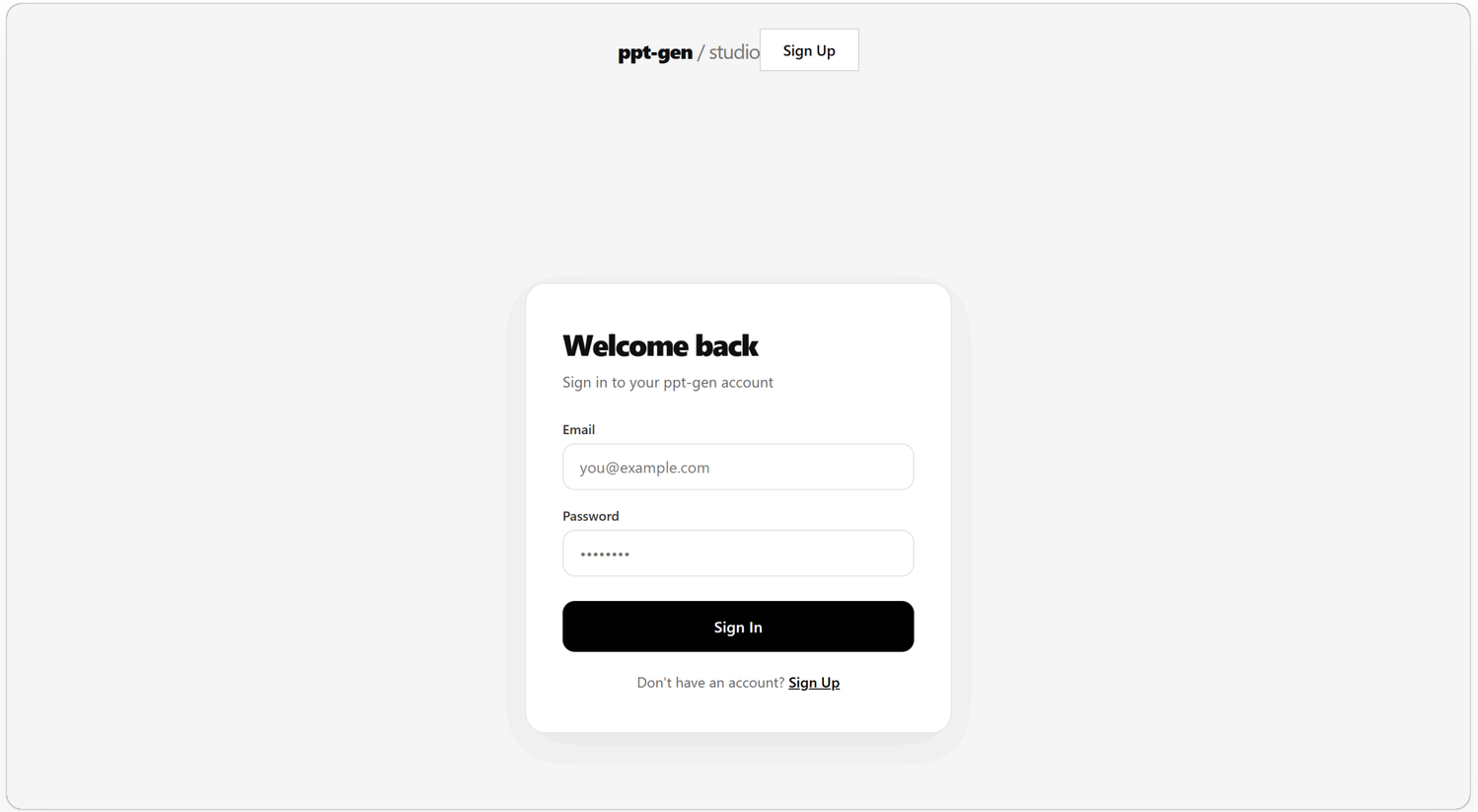


Figure 2: Secure sign-in dialog with validation feedback (/login)

Backend JWT Authentication

Security Architecture

- **Stateless Token Auth:** Client stores signed JWT bearer token securely in browser storage.
- **Automatic Route Guard:** <ProtectedRoute> wrapper instantly redirects unauthenticated users to /login.
- **Input Normalization:** Trimmed strings and lowercased email sanitization prevent case sensitivity bugs.
- **BCrypt Hashing:** Passwords protected with industry-standard adaptive salted hashes on Render backend.
- **One-Click Switching:** Seamless navigation between Login and Signup modals with unified error messaging.

Studio Projects Dashboard

SHOWCASE 3

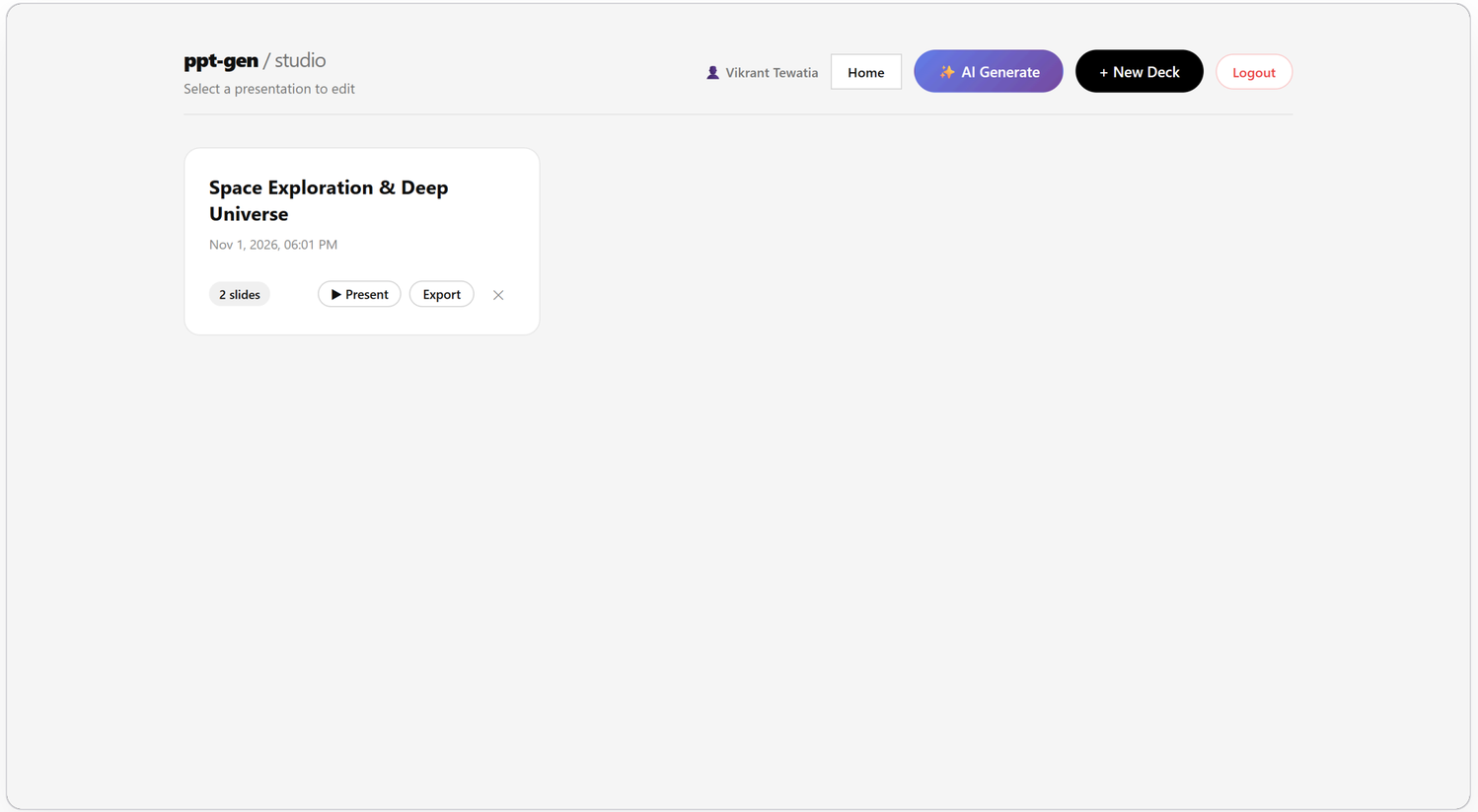


Figure 3: Authenticated user dashboard with multi-deck grid (/projects)

MongoDB Atlas Cloud Sync

Dashboard Controls

- **User Profile Header:** Displays authenticated user name with clean logout and home navigation pills.
- ✨ **AI Generate Trigger:** High-visibility purple accent button opening prompt engineering generator.
- **Deck Cards Grid:** Dynamic cards showing presentation title, slide count badge, and formatted timestamp.
- ▶ **Present Button:** Direct trigger on each card allowing instant presentation launch without editor loading.
- **Direct Card Export:** Allows immediate compilation into PDF or PowerPoint directly from dashboard overview.

Intelligent Gemini AI Generation Modal

SHOWCASE 4

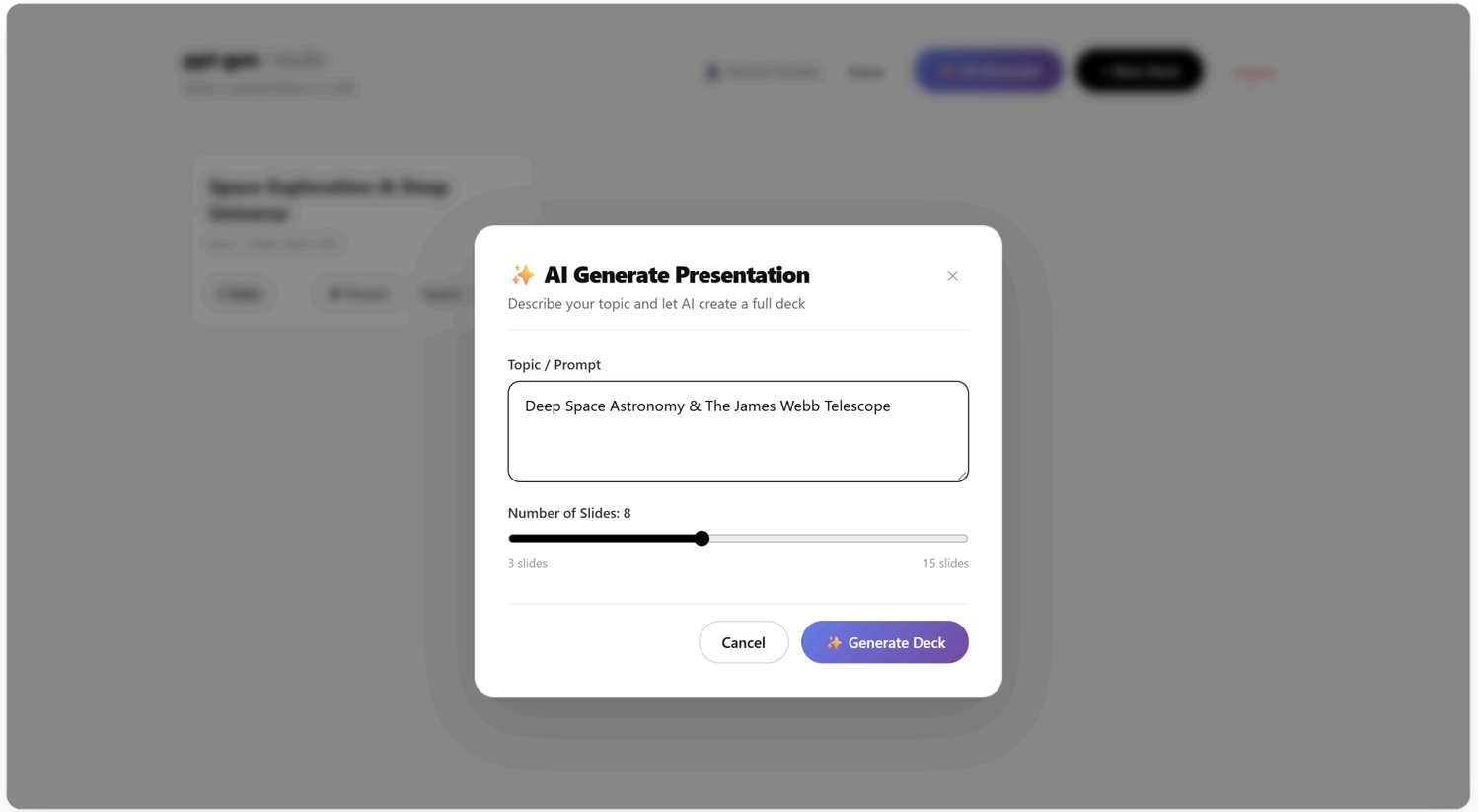


Figure 4: AI presentation creation modal with topic input and length slider

Google Gemini 3 Flash Preview

Generation Capabilities

- **Natural Language Input:** Accepts high-level topic titles, detailed technical prompts, or domain summaries.
- **Dynamic Slide Range:** Slider allows users to dial exact slide counts from 3 up to 15 structured slides.
- **Sub-2s LLM Response:** Powered by Gemini 3 Flash with strict JSON Schema output enforcement.
- **Zero Slop Guarantee:** Enforces factual bullets, technical concepts, real metrics, and zero marketing filler.
- **Loading State HUD:** Clean animated spinner communicating concurrent text generation and Unsplash image fetching.

16:9 Presentation Editor & Unsplash Visuals

SHOWCASE 5

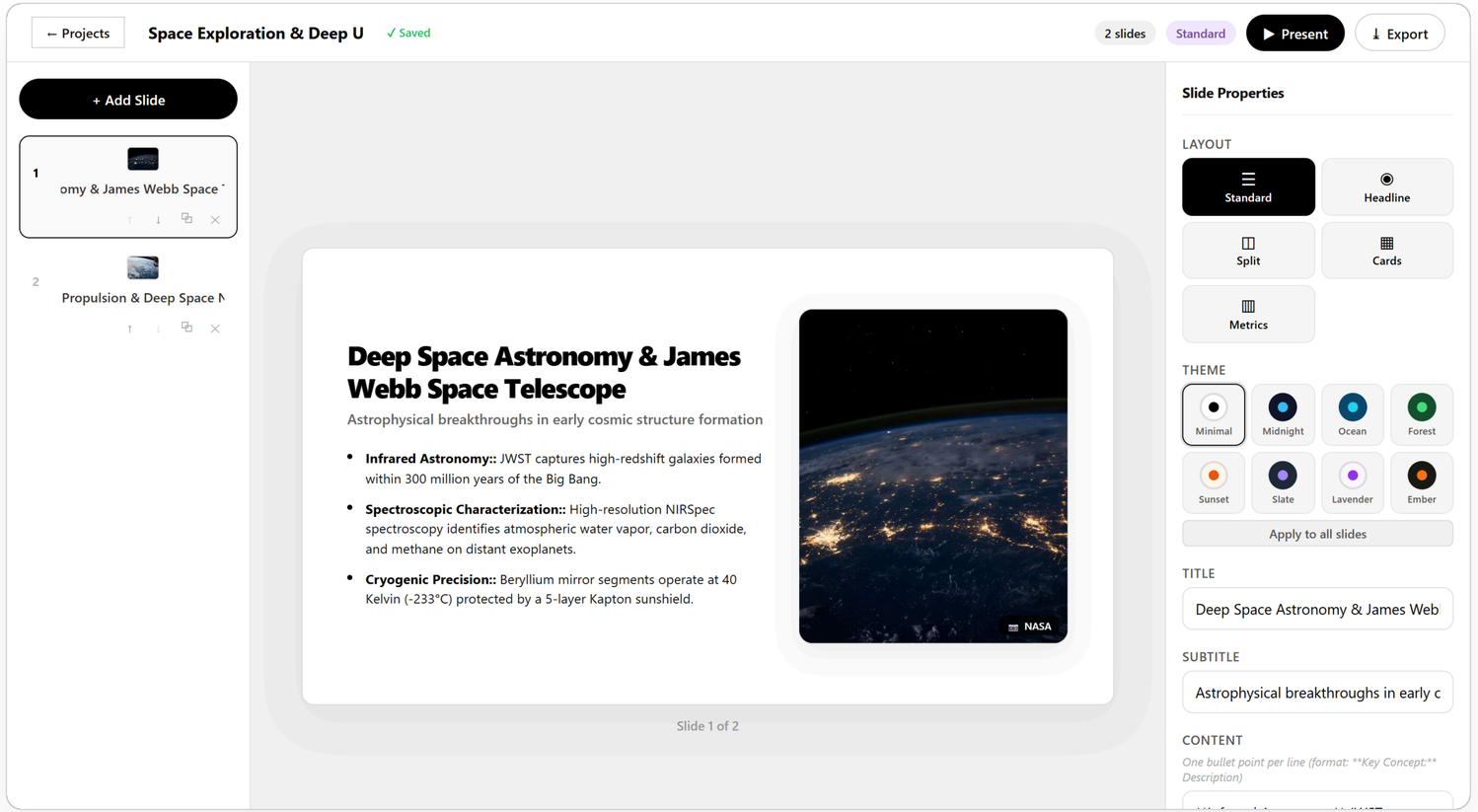


Figure 5: 16:9 slide canvas with 42% Unsplash photo card and photographer credit

Distraction-Free Inline Editor

Canvas Features

- **42% Split Image Card:** Curated high-resolution Unsplash photo rendered in dedicated right-column card.
- **Photographer Attribution:** Official badge linking photographer's Unsplash profile directly.
- **Cleaned UI (No Copilot):** Removed sidebar copilot distraction to focus completely on slide content and structure.
- **Markdown Bold Parsing:** Automatically renders **Key Concept:** prefixes in bold black contrast.
- **Aspect-Locked Geometry:** Enforces exact 16:9 ratio, completely eliminating broken margins across displays.

Phase 4 Fullscreen Present Mode

SHOWCASE 6

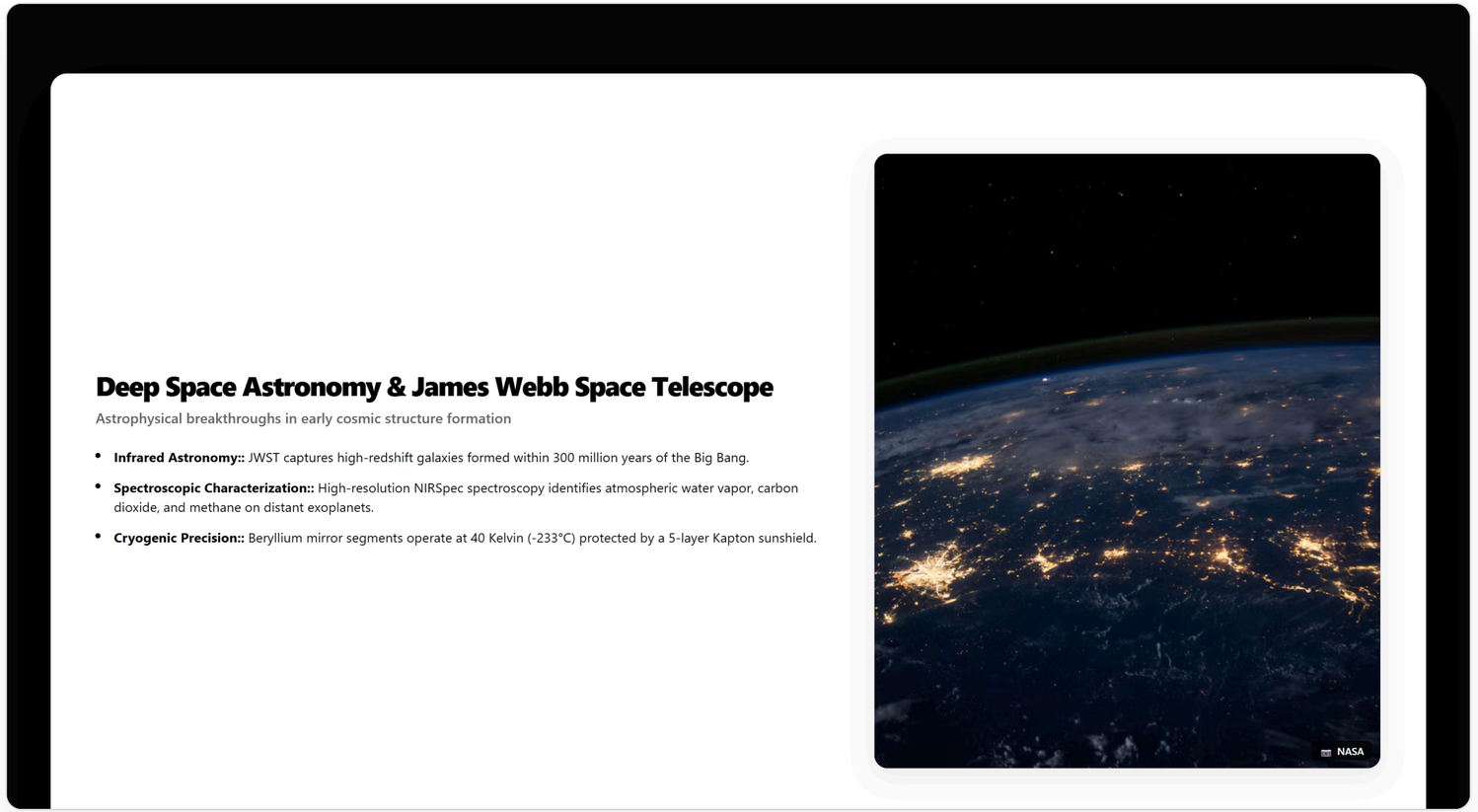


Figure 6: Fullscreen cinematic presentation engine with HUD controls and progress bar

Phase 4 Milestone Feature

Presentation Controls

- **Keyboard Shortcuts:** Spacebar or Right Arrow to advance, Left Arrow to reverse, Esc to exit.
- **Native Fullscreen Toggle:** Press F key or click HUD icon for true browser fullscreen presentation.
- **Automated Autoplay:** Press P key to cycle through presentation automatically at 5-second intervals.
- **Auto-Hiding Floating HUD:** Controls auto-fade during mouse inactivity to deliver an immersion-first experience.
- **Bottom Progress Bar:** Real-time linear progress bar communicating visual pacing through the deck.

Streamlined 5-Format Export Engine

SHOWCASE 7

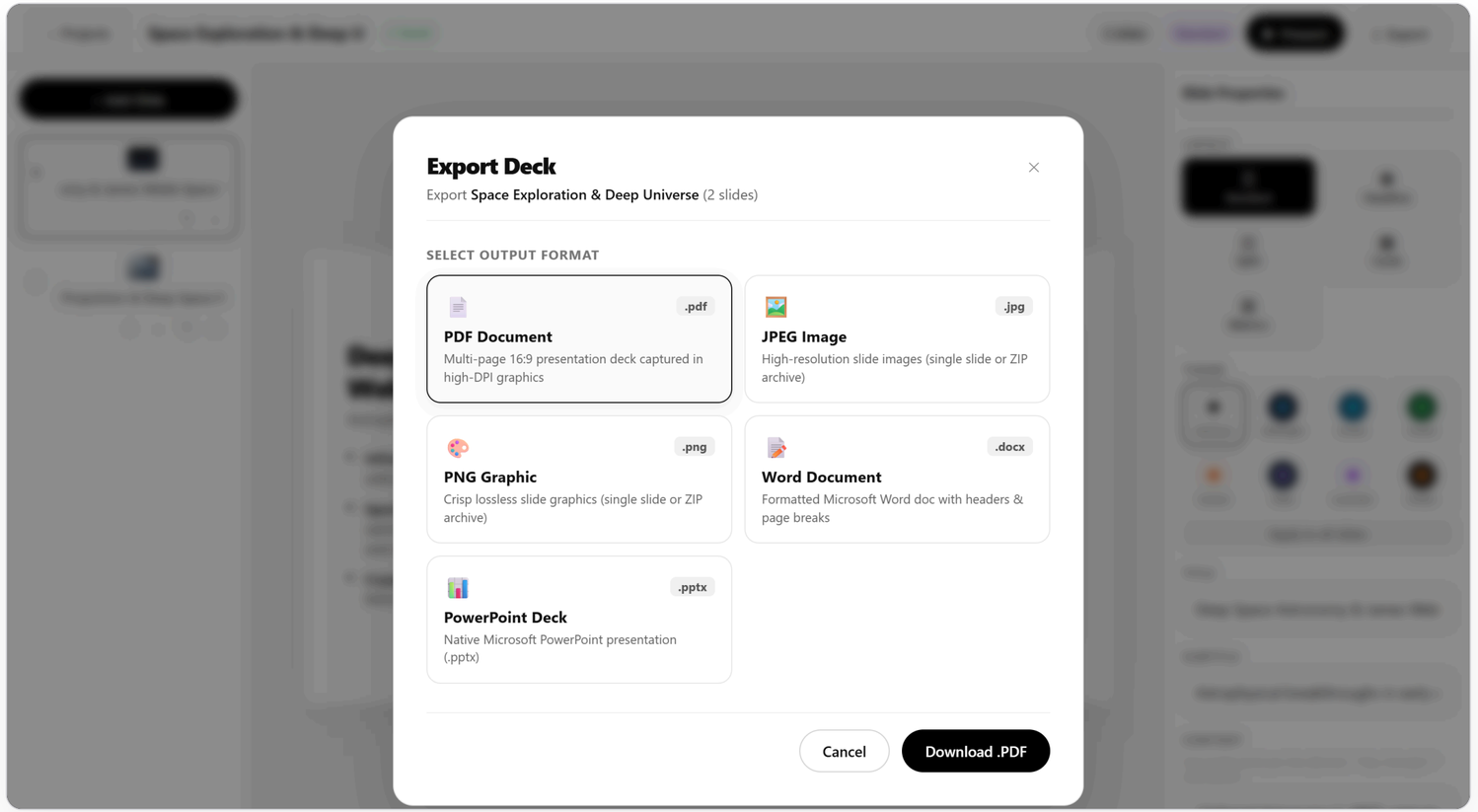


Figure 7: Streamlined export modal restricted strictly to 5 production formats

Client-Side Document Compilers

5 Production Formats

- **1. PDF Document:** Multi-page landscape PDF compiled at 2x High-DPI resolution via jsPDF and html2canvas.
- **2. JPEG Image Deck:** High-compression rasterized image export for quick sharing and mobile viewing.
- **3. PNG Graphic Deck:** Lossless crisp graphic format with transparency preservation.
- **4. Microsoft Word (.docx):** Structured document generated using client-side docx Packer.
- **5. PowerPoint Deck (.pptx):** Native widescreen 16:9 slides built via pptxgenjs for offline PowerPoint editing.

Engineering Challenges Faced & Solved

PROBLEM SOLVING

Technical Hurdle	Architectural Solution Implemented
Google Gemini Model Deprecation (gemini-2.0-flash 404)	Migrated to gemini-3-flash-preview with responseMimeType: 'application/json', reducing generation latency from ~90s down to ~1.9s.
Sequential Image Fetching Bottlenecked Deck Creation	Engineered a parallel Promise.all Unsplash pipeline executing all slide image queries concurrently without blocking Express event loop.
Washed-out / Invisible Slide Images	Overhauled SlideRenderer.jsx to render true <img> elements inside prominent 42% split layout cards with dark hero backdrops.
GitHub Secret Scanning Blocked Git Push	Reset repository tree, updated render.yaml with sync: false for all secrets, and enforced strict .gitignore rules.
Render Free Tier Cold Start & URL Normalization	Implemented defensive URL trimming in src/utils/api.js guaranteeing accurate routing regardless of trailing slash or missing /api.
Vercel SPA Client-Side Direct Routing 404s	Configured vercel.json rewrites routing all dynamic paths back to /index.html, preserving React Router state on reload.

Production Cloud Deployment & Verification

DEVOPS

100%	~1.9s	5 / 5	0 KB
API HEALTH UPTIME	GEMINI 3 LATENCY	EXPORT FORMATS LIVE	SERVER DISK LEAK

Component	Host / Service	Live URL / Endpoint	Status
Frontend Application	Vercel Global Edge Network	https://ppt-gen-two.vercel.app/	ACTIVE
Backend Web Service	Render Cloud Platform (Node.js)	https://ppt-gen-backend-doal.onrender.com/api/health	ACTIVE
Database Cluster	MongoDB Atlas (AWS Cloud)	Cluster0 (mongodb+srv://...)	CONNECTED
Git Repository	GitHub (Main Branch)	https://github.com/Vikrant153-gif/ppt-gen.git	SYNCHRONIZED

Conclusion & Next Milestones

MILESTONES

Progress Report II Accomplishments

PPT-GEN has successfully transitioned from a client-side prototype into a **production-grade, cloud-deployed full-stack platform**. By combining Google Gemini 3 Flash structured prompting, Unsplash concurrent HD media discovery, Phase 4 Fullscreen Present Mode, and 5 client-side document compilers, PPT-GEN delivers end-to-end presentation authoring in seconds without layout degradation.

Upcoming Milestones (Final Phase)

- **Dynamic Audio Narration (Text-to-Speech)**
  Integrating Web Speech API for automated slide voiceovers during present mode playback.
- **Interactive Chart & Graph Schemas**
  Expanding layout schemas with SVG bar, line, and radar chart generation directly from Gemini data.
- **Multi-User Real-Time Collaboration**
  WebSocket-based synchronized room editing allowing teams to co-author slide decks simultaneously.

Thank You. Demonstrations & Questions Welcome!